Analisis del circuito por superposición

28 | 10 | 26

Circuito 1. Apagando $F_0(t)$

$$x(t) = x_1(t) + x_2(t)$$

$$x(t) = C_p \frac{dF_{S1}(t)}{dt}$$

$$x_1(t) = C_s \frac{d[F(t) - F_{S1}(t)}{dt}$$

$$x_2(t) = \frac{F(t) - F_{S1}(t)}{R}$$

$$C_p \frac{dF_{S1}(t)}{dt} = C_s \frac{d[F(t) - F_{S1}(t)]}{dt} + \frac{F(t) - F_{S1}(t)}{R}$$

$$C_p s F_{S1}(S) = C_s S [F(S) - F_{S1}(S)] + \frac{F(S) - F_{S1}(S)}{R}$$

$$C_p s F_{S1}(S) = C_s s F(S) - C_s s F_{S1}(S) + \frac{F(S)}{R} - \frac{F_S(S)}{R}$$ Transformada de Laplace

$$\left(C_p s + C_s s + \frac{1}{R}\right) F_{S1}(S) = \left(C_s s + \frac{1}{R}\right) F(S)$$

$$\frac{R(C_p + C_s)s + 1}{R} F_{S1}(S) = \frac{C_s R s + 1}{R} F(S)$$

$$F_{S1}(S) = \frac{\frac{C_s R s + 1}{R} F(S)}{\frac{R(C_p + C_s)s + 1}{R}} = \frac{C_s R s + 1}{R(C_p + C_s)s + 1} F(S)$$

Circuito 2. Apagando $F(t)$

$$-\alpha F(t) = R \cdot x(t) + \frac{1}{C_p + C_s} \int x(t)\, dt$$

$$F_{S2}(t) = \frac{1}{C_p + C_s} \int x(t)\, dt$$

$$-\alpha F(S) = R x(S) + \frac{x(S)}{(C_p + C_s)s}$$

$$F_{S2}(S) = \frac{x(S)}{(C_p + C_s)s}$$

$$F(S) = -\frac{x(S)}{\alpha} \frac{R(C_p + C_s)s + 1}{C_p + C_s)s}$$

$$\frac{F_{S2}(S)}{F(S)} = \frac{\frac{x(S)}{(C_p + C_s)s}}{\frac{x(S)}{\alpha} \frac{R(C_p + C_s)s + 1}{(C_p + C_s)s}} = -\frac{\alpha}{R(C_p + C_s)s + 1}$$

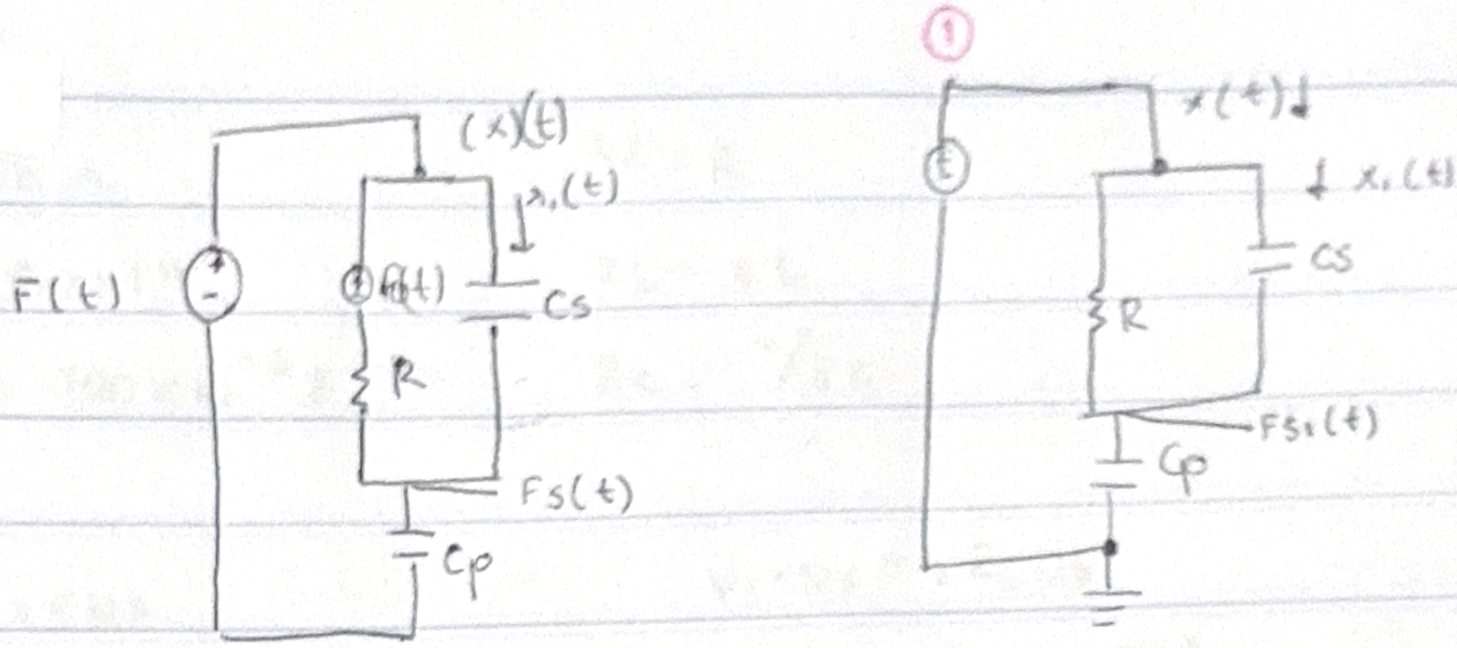
$$= F_{S2}(S) = \frac{\alpha F(S)}{R(C_p + C_s)s + 1}$$

Función de transferencia

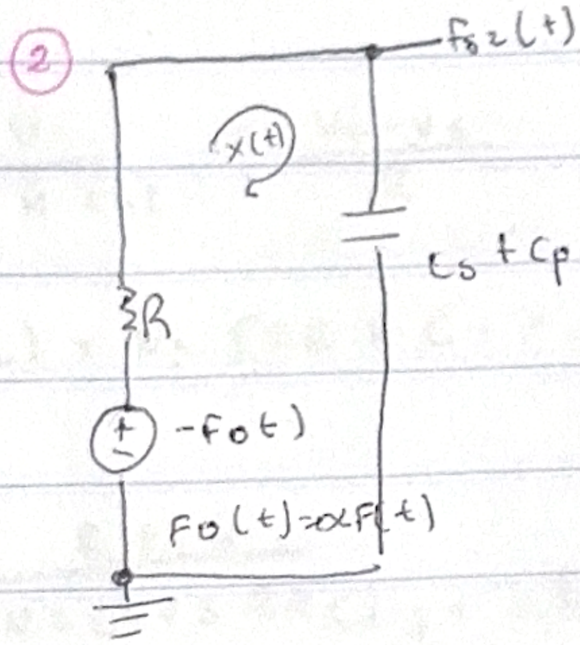
$$F_S(s) = F_{S1}(s) + F_{S2}(s) = \frac{C_S R_S + 1}{R((C_p + C_S)S + 1} \qquad F(s) - \frac{\alpha F(s)}{R((C_p + C_S)S + 1}$$

$$F_S(s) = \frac{C_S R_S + 1 - \alpha}{R(C_p + C_S)S + 1} F(s)$$

$$\frac{F_S(s)}{F(s)} = \frac{C_S R_S + 1 - \alpha}{R(C_p + C_S) + 1}$$


x(t)
x1(t)
F(t)
-
+
Cs
R
Fs(t)
Cp

1
x(t)
x1(t)
Cs
R
Fs1(t)
Cp


2
fs2(t)
x(t)
Cs + Cp
R
-Fo(t)
Fo(t) = αF(t)